

Use Case Diagram

PRESENTED BY : -

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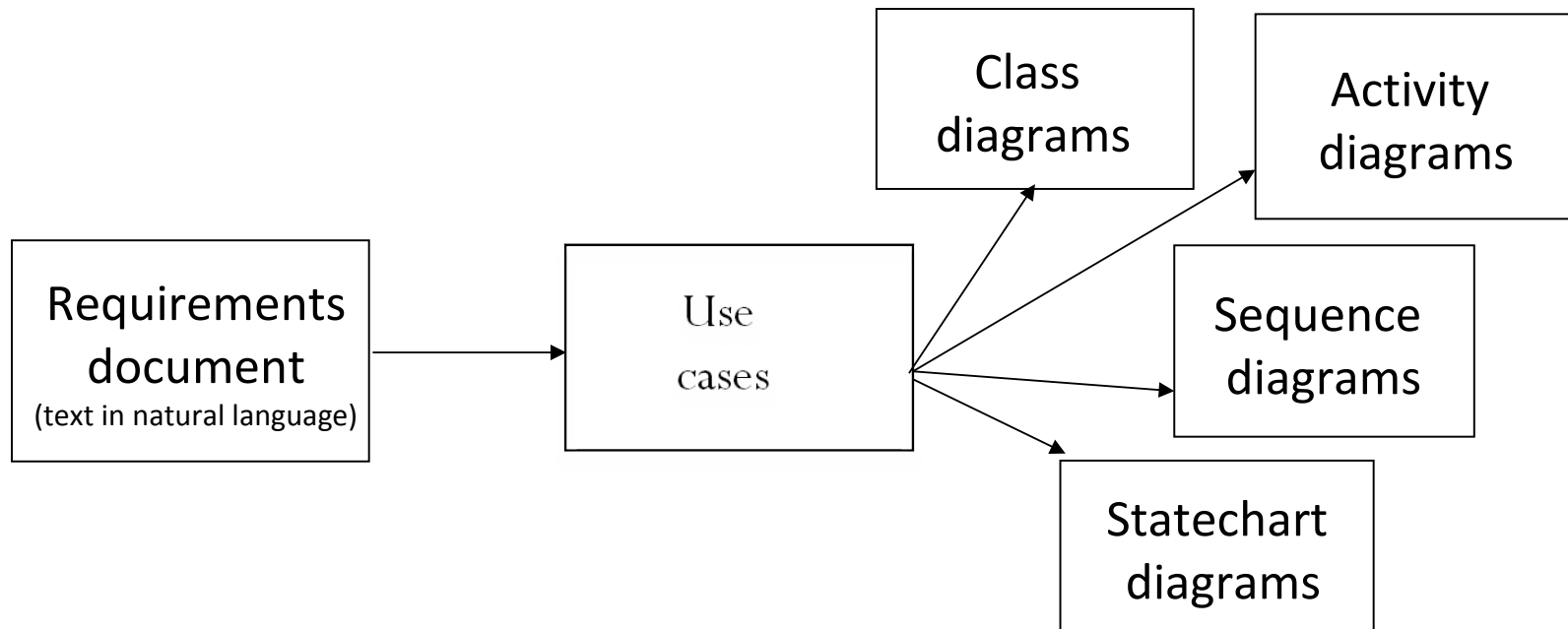
- What is Use-Case Diagram
- Purpose

Introduction

Use-cases are descriptions of the functionality of a system from a user perspective.

- ❖ Depict the behaviour of the system, as it appears to an outside user.
- ❖ Describe the functionality and users (actors) of the system.
- ❖ Show the relationships between the actors that use the system, the use cases (functionality) they use, and the relationship between different use cases.
- ❖ Document the scope of the system.
- ❖ Illustrate the developer's understanding of the user's requirements.

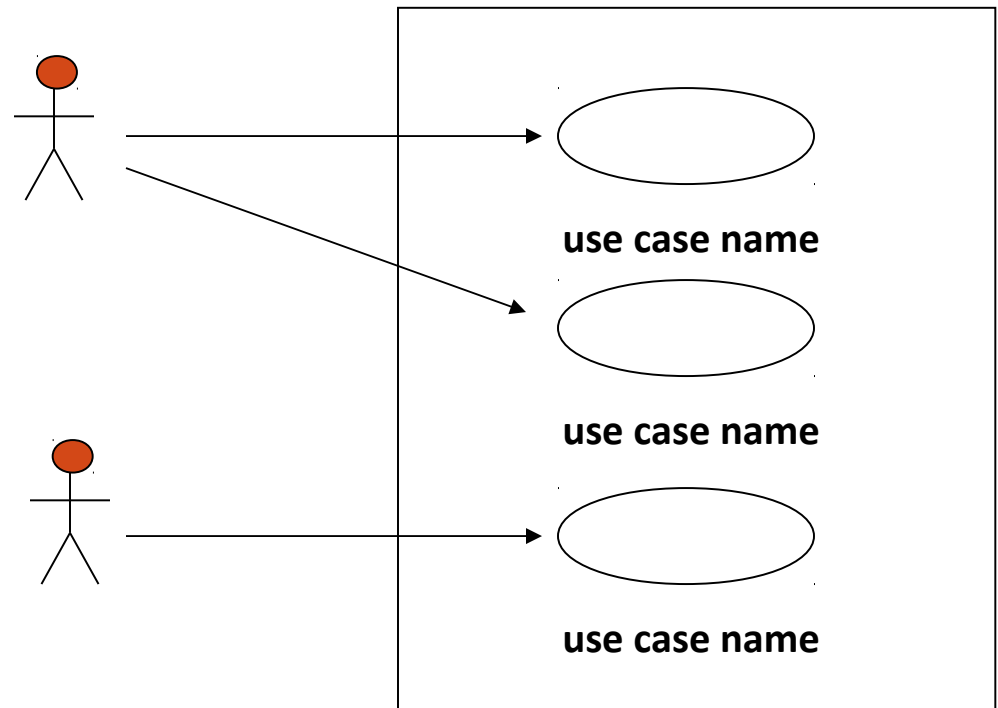
Use Case Diagram, purpose



- Use case models are developed at different levels of abstraction
 - system, system component, or a class.
- Use case modelling is an iterative and incremental process.
 - If user requirements change, the changes should be made in all the affected documents.

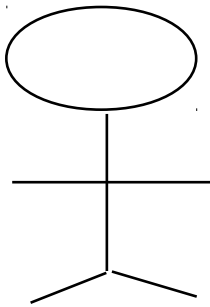
Use Case diagrams, basic UML notation

- Use Case: A Use Case is a description of a set of interactions between a user and the system.
- Components of use case diagram:
 - Actor
 - Use case
 - System boundary
 - Relationship

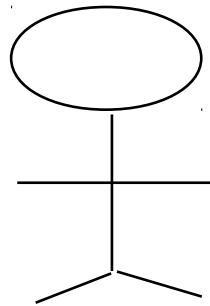


ACTOR

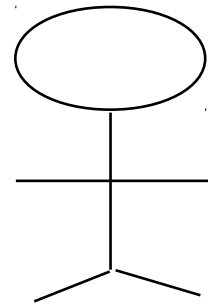
- An actor is some one or something that must interact with the system under development
- Actors can be human or automated systems.
- Actors are not part of the system.
- UML notation for actor is stickman, shown below.



Student



Faculty



Employee

Primary and Secondary Actors

- Primary Actor
 - Acts on the system
 - Initiates an interaction with the system
 - Uses the system to fulfill his/her goal
 - Events Something we don't have control over
- Secondary Actor
 - Is acted on/invoked/used by the system
 - Helps the system to fulfill its goal
 - Something the system uses to get its job done

External Hardware And Other Systems

- **External Hardware**
 - It is the hardware device
 - It is the part of an application
 - Uses the system to fulfill his/her goal
- **Other Systems**
 - Is acted on/invoked/used by the system
 - Helps the system to fulfill its goal
 - The system with which system interacts

USE CASE

What is USE case?

- A use case is a pattern of behavior, the system exhibits
- The use cases are **sequence of actions** that the user takes on a system to get particular target
- USE CASE is dialogue between an actor and the system.

- **Examples:**



Add a course

System Boundary

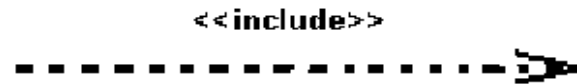
- It is shown as a rectangle.
- It helps to identify what is external versus internal, and what the responsibilities of the system are.
- The external environment is represented only by actors.



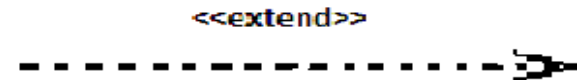
Relationship

- Relationship is an association between use case and actor.
- There are several Use Case relationships:

- Include



- Extend



- Association



- Dependency

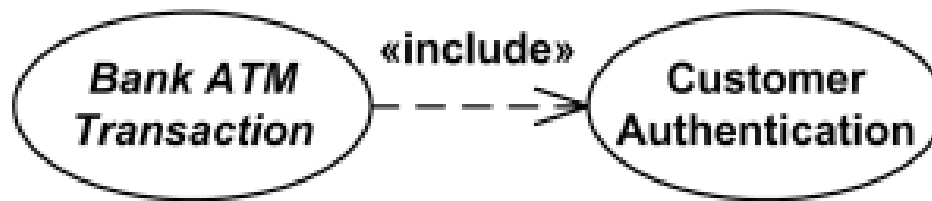


- Generalization



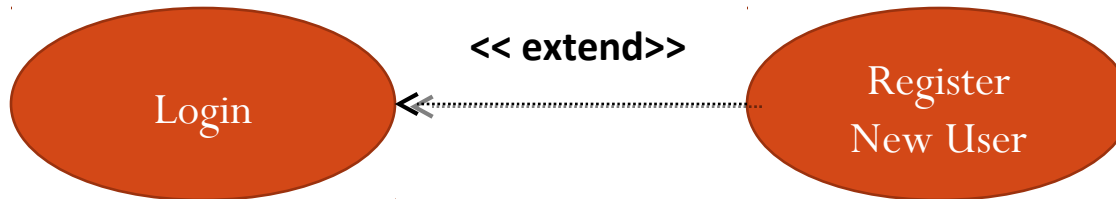
Include Relationship

- Include relationships insert additional behavior into a base use case
- use cases that are included as parts of other use cases. Enable to factor common behavior.
- They are shown as a dotted line with an open arrow and the key word <<include>>

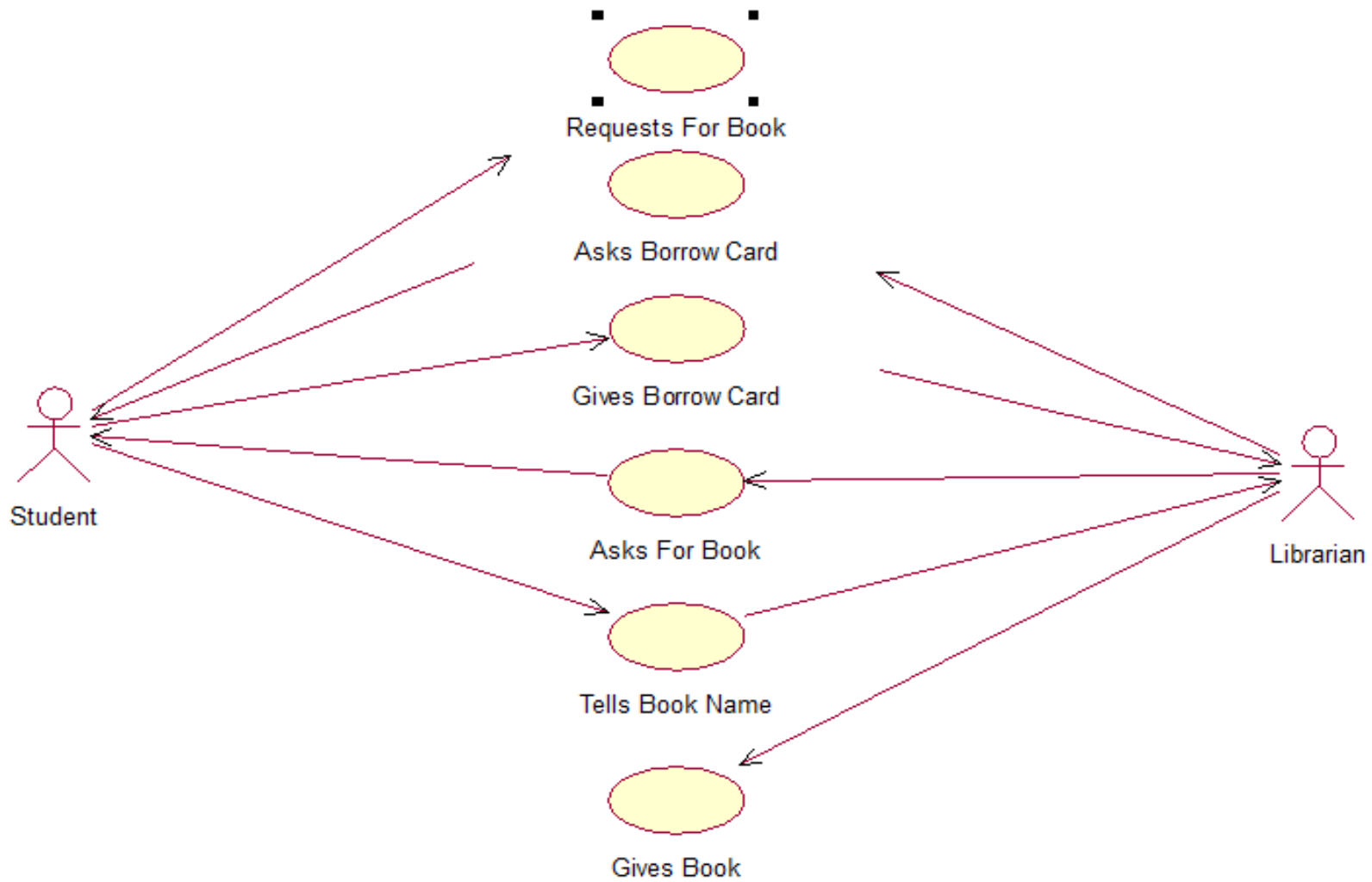


Extend Relationship

- The extended relationship is used to indicate that use case completely consists of the behavior of another use case at one or specific point
- use cases that extend the behavior of other core use cases. Enable to factor variants
- It is shown as a dotted line with an arrow point and labeled <<extend>>



USECASE DIAGRAM FOR LIBRARY MANAGEMENT



THANK YOU